

A Review of Deep Learning Application in Cryptography

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Abstract—Deep Learning-based cryptosystem is an emerging field of cryptography with less computational overhead and improved accuracy. 'Deep Learning in Cryptography' is an attempt to study various Deep Learning applications in the field of cryptography. Transforming plaintext into ciphertext requires complex mathematical calculations and an algorithmic approach for making the system more difficult and secure. This additional overhead can be minimized using the concept of data science such as Artificial Intelligence, Machine Learning and Deep Learning. The complex network structure in neural networks makes the system more secure. Deep Learning-based model predictions can be applicable for malware classification, intrusion detection, and multichannel attacks. The efficient feature extraction process can produce an accurate result from the given set of input data. The featured input structure passes through various layered architecture and produces better classification output with good accuracy.

Index Terms—Cryptography, Privacy Preserving, Deep Learning, Intrusion Detection, Artificial Intelligence.

I. INTRODUCTION

Artificial intelligence is an emerging field of computer science capable of performing intelligence tasks as like as human brain. Machine learning and deep learning are a broad category of Artificial intelligence. Machine learning requires structured data as input while deep learning requires layers of a neural network for processing inputs. The Deep Learning approach can be used for model prediction such as image classification, speech recognition, natural language processing and malware detection. The problems which are difficult for human and statistical standards can be solvable by Artificial Neural Networks. Deep learning is the subfield of machine learning and also belongs to the family of Artificial Neural Networks. Artificial neural network receives mathematically designed patterns and images as inputs for processing and produces output in the form of prediction or classification. The observed inputs pass through multiple layers of neural networks and until the attainment of the final output. The applications of stochastic algorithms such as artificial neural network algorithms can be used for encryption and cryptanalysis in the field of neural cryptography. Artificial neural network provides various approaches for data encryption and testing the strength of encryption data. Deep Learning models can

be employed in the field of image encryption using CycleGAN network[12].

Deep learning approaches can be applied for the detection of multichannel attacks in social networks. Deep learning plays an important role in cryptography. Deep Learning can be applied in the field of cryptography for cryptographic applications such as encryption, malware detection and classification. Deep learning and cryptanalysis are more common than of cryptography and DL. DNA cryptography is the branch of cryptography which employs deep learning approaches for key generation using genetic algorithms with a neural network algorithm [1]. Nowadays, cyber attacks become difficult to detect in a variety of fields such as industry, national defense, and healthcare. Deep learning-based intrusion detection is an alternative solution instead of traditional intrusion detection systems[11] through a deep neural network. Attackers can be able to pretend to be normal users by bypassing known signatures. Deep learning is an alternative to solving these issues. Deep learning extracts intrusion features through training input data. Known signature based attack detection and anomaly detection based on normal use patterns are the two types of intrusion detection system.

The neural network model composed of artificial neurons and nodes in a circuit manner. Artificial Neural Networks and Deep Learning methods are powerful AI techniques for solving complex problems. In Deep Learning approach, the input data passes through various complicated hidden layers and produces a target output. The meaningful results can be produced by efficient feature extraction methods introduced in Deep Learning algorithms. Nowadays, Cryptography and cryptanalysis are the emerging trends of deep learning. Deep learning based cryptosystem, attacks classification, privacy preservation[10] and encryption algorithm analysis are the various research areas in deep learning based cryptosystem. Deep learning based cryptanalysis can be applied in the field of light weight block ciphers.

II. RELATED WORKS

In [3] introduced chaotic based algorithms to provide secure storage and transmission of images. Encryption algorithm is controlled by the newly introduced heterogeneous

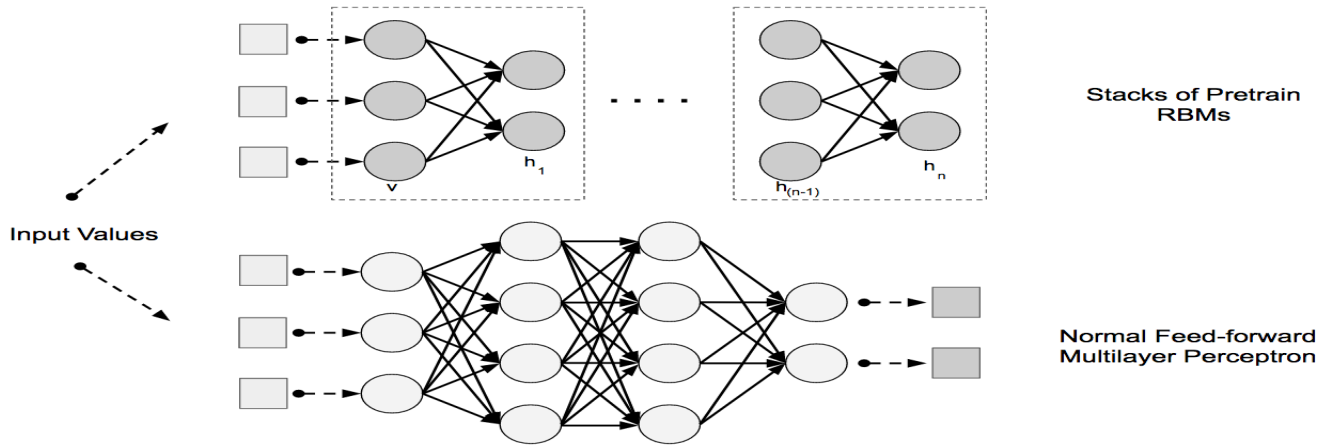


Fig. 1. Deep Learning Layered Architecture.

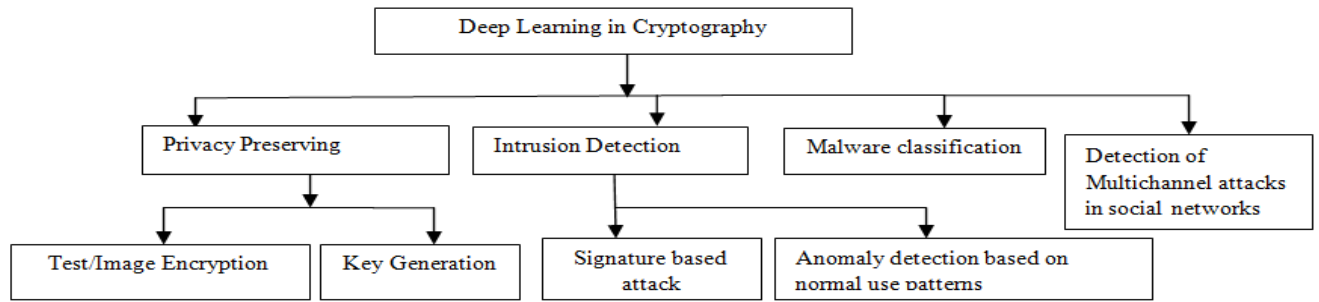


Fig. 2. An Overview of Deep Learning in Cryptography.

chaotic neural network generator. Chaotic generator as input for the operations of DNA based bit substitution and permutations. Degree of randomness is tested based on auto correlation, cross correlation, I/O balance and NIST test. The generation of chaotic sequence depended on the number of neurons in the output layer. Logistic Chaotic Map, Piece Wise Linear Chaotic Map and Logistic-Tent system are introduced in Neural Network layers. The proposed system performs I/O balance on 10 orders best. Histogram analysis is performed by taking 100 encrypted images as input and obtaining the average pixel intensity of 127.46. The ideal value was 127.5. The system provides a better entropy value for encrypted images. The entropy value is compared with other similar works and achieved a high cryptographic quality.

In [2] proposed an integrative approach for DNA cryptography with Deep Learning. Natural selection process of the genetic algorithm for key generation is mixed with Needleman Wunsch algorithm which is capable of finding dissimilarity in DNA strands. The deep learning algorithm generates a complex cryptosystem based on biological operations and human behaviour by using the input key and DNA cryptography [1]. This method increases the security and reduces the mathematical complexity for the formulation of the algorithm. The DNA computing follows symmetric encryption and decryption. The encoded nucleotides are

encrypted with a key which is generated through genetic algorithms. The complexity of the algorithm makes the process more secure.

In [4] introduced a deep learning based cryptosystem following auto-encoder neural network for symmetric encryption and decryption. Replicative capability of the auto-encoder forms the key of the sequence of seed values for a symmetric system. This is a pseudorandom process. After the fine tuning process, the sender and receiver can exchange keys. This system proposed an efficient and secure cryptographic algorithm. A sparse auto-encoder is used for encryption and decryption and the learning process is done by a key generator. This encryption process can be applied to multimedia files in the future. In [5] introduced a novel network intrusion detection system based on the Deep Learning model which offers more feasibility and sustainability. This application uses asymmetric stacked deep autoencoder for unsupervised feature learning which is combined with a random forest algorithm. This system was implemented with GPU based TensorFlow, and experimented using the KDD data set. This model offers average accuracy of 95% which is greater than the performance and accuracy of existing techniques.

TABLE I
VARIOUS DEEP LEARNING METHODS IN CRYPTOGRAPHY

Article	Outline of the study	Focused Domain	Learning Architectures
Aldweesh et al.(2019) [8]	Classification for Deep learning based IDS, Compared the performance of the proposed methods through descriptive analysis.	Intrusion Detection(IDS)	AE,RBM,DBN, RNN,CNN
Kalsi, S et al.(2018) [2]	DNA cryptography with Deep Learning for encryption and decryption	Data security	DNN
Tran et al.(2020) [6]	Data privacy among users is implemented, Introduced a secure decentralized training framework.	Privacy preserving	CNN and LSTM centralize
Warit Sirichotedumrong et.al(2019)[9]	A novel approach to pixel based image encryption without key management. Provides protection for visual information.	Privacy preserving	Deep Residual network , DNN
Shone et.al (2019) [5]	Unsupervised feature learning with a non symmetric auto-deep encoder is implemented, which improves the accuracy and performance of the existing techniques.	Network Intrusion Detection.	Deep Auto-Encoders, RF-ensemble learning method.
Francisco Quinga-Socasi et.al (2020)[4]	Symmetric encryption key combines RNN and Deep Learning approaches. The system is evaluated with different cryptographic algorithms.	Symmetric Key Cryptography based on Deep Learning Technique.	ANN
Gururaj Maddodi et.al(2018) [3]	Secure transmission and storage of images based on chaotic encryption algorithm and Chaotic pseudo-random generator based on neural network is introduced.	Deep Learning based Symmetric key cryptosystem.	ANN

In [6] introduces a new well-organized deep learning model framework named Secure Decentralized Training Framework (SDTF) for privacy-preserving. This model provides privacy for local data and also capable of working decentralized network setting without a trusted third-party server. This system proposes an Efficient Secure Sum Protocol (ESSP) which is capable of enabling a large group of parties to jointly compute a sum of private inputs and cryptographically strong. This protocol assurance high privacy with lesser communication costs. The framework is based on two input datasets such as balance class image datasets (MNIST) and unbalance class text data set (UCI SMS spam) and obtained a very high-accuracy of 97

In [8] discusses the role of deep learning in intrusion detection and provides a novel categorical approach of the deep learning-based IDSs with respect to different features, including input data, detection, deployment, and evaluation strategies. The features are again classified according to different criteria. The various intrusion detection methods are compared in terms of feature learning, classification technique, testing methodology and efficiency. This study ensures that Deep Learning-based anomaly detection

achieves a high accuracy with the use of different deep learning architectures using benchmark datasets, KDD99 and NSLKDD.

In [9] proposed encrypting images by using different encryption keys. All clients are able to utilize independent keys for training and testing a model. Hence, there is no need to use key management techniques. The independent key performs high classification performance and vigorous against DNN based attacks. Data augmentation is also introduced in the encrypted domain. The system evaluates its performance in terms of accuracy and robustness against DNN-based attacks.

This study provides a theoretical framework describing Deep Learning in the field of cryptography. The neural network approach can be applicable for various cryptographic techniques such as encryption, cryptanalysis, steganography, intrusion detection, malware classification [7], social network attacks detection and privacy preserving. The integrative approach of deep learning methods and DNA cryptography plays an important role in data security.

III. CONCLUSION

The objective of this study is to evaluate the various applications of Deep Learning in the field of cryptography. This study provides an integrative research scope in the field of Deep Learning based cryptosystem. As comparing with traditional methods, the Deep Learning approach reduces the mathematical complexity of algorithms and computational cost. TensorFlow supports CPU and GPU based computation which provides better accuracy. The Deep Learning technique overcomes the security problems in traditional cipher by using Deep Neural Network System approach.

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